CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 23: Graphical Representation (Histograms and Ogives)

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Mode from a histogram

1. Draw a histogram for the marks of 164 students given below, and use it to find the mode.

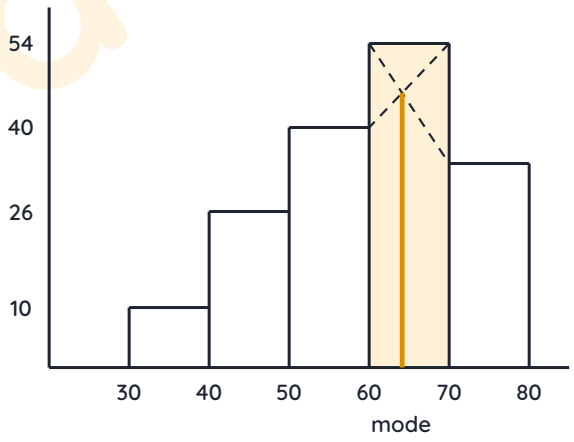
ICSE 2025, 3 marks

Marks	30–40	40–50	50–60	60–70	70–80
Number of students	10	26	40	54	34

SOLUTION

Draw a rectangle on each class with height equal to its frequency. The tallest rectangle (60–70) is the modal class.

Join the top corners of the tallest rectangle to the nearer top corners of the rectangles on either side, as shown. From the point where the two lines cross, drop a perpendicular to the marks axis.



The perpendicular meets the axis at about 64.

Check with the formula: mode $= 60 + \dfrac{54 - 40}{2(54) - 40 - 34} \times 10 = 60 + \dfrac{140}{34} \approx 64.1$

Answer: mode ≈ 64 marks

2. A histogram shows the heights of the plants in a school garden. Its rectangles stand on the classes 30–40, 40–50, 50–60, 60–70, 70–80, 80–90 and 90–100 (in cm) and have heights 4, 2, 8, 12, 6, 3 and 4. (a) Make the frequency table. (b) State the modal class. (c) Find the mode. (d) How many plants are between 80 cm and 90 cm tall? ICSE 2024, 5 marks

SOLUTION

(a) Read the height of each rectangle:

Height (cm)	30–40	40–50	50–60	60–70	70–80	80–90	90–100
Number of plants	4	2	8	12	6	3	4

(b) The tallest rectangle is on 60–70. **Modal class: 60–70.**

(c) Join the top corners of the tallest rectangle to the nearer top corners of its neighbours. The lines cross above 64 on the axis. Check: $60 + \frac{12 - 8}{24 - 8 - 6} \times 10 = 64$. **Mode $= 64$ cm.**

(d) The rectangle on 80–90 has height 3. **3 plants.**

3. The daily wages of 60 workers are given below. Draw a histogram and use it to find the mode. ICSE 2023, 3 marks

Daily wage (₹)	200–300	300–400	400–500	500–600	600–700	700–800
Number of workers	6	10	14	16	10	4

SOLUTION

Draw the rectangles. The tallest is on 500–600. Join its top corners to the nearer top corners of the rectangles on either side, and drop a perpendicular from the crossing point to the wage axis.

Check with the formula: mode $= 500 + \frac{16 - 14}{32 - 14 - 10} \times 100 = 500 + 25$

Answer: mode $=$ ₹525

4. Which of these cannot be found graphically for a grouped frequency distribution? (a) median (b) mode (c) quartiles (d) mean ICSE 2023, 1 mark

SOLUTION

The mode is read from a histogram, and the median and quartiles from an ogive. The mean has to be calculated.

Answer: (d)

5. Draw a histogram for the weekly wages below and estimate the mode from it. ICSE 2020, 4 marks

Weekly wages (₹)	3000–4000	4000–5000	5000–6000	6000–7000	7000–8000	8000–9000	9000–10000
Number of people	4	9	18	6	7	2	4

SOLUTION

The tallest rectangle is on 5000–6000. Join its top corners to the nearer top corners of the rectangles on either side, and drop a perpendicular from the crossing point to the wages axis.

Check with the formula: mode $= 5000 + \frac{18-9}{36-9-6} \times 1000 = 5000 + \frac{9000}{21} \approx 5429$

Answer: mode is about ₹5430

6. The runs scored by 50 batsmen are given below. Draw a histogram and estimate the mode. ICSE 2018, 4 marks

Runs scored	3000–4000	4000–5000	5000–6000	6000–7000	7000–8000	8000–9000	9000–10000
Number of batsmen	4	18	9	6	7	2	4

SOLUTION

The tallest rectangle is on 4000–5000. Join its top corners to the nearer top corners of the rectangles on either side, and drop a perpendicular from the crossing point to the runs axis.

Check with the formula: mode $= 4000 + \frac{18-4}{36-4-9} \times 1000 = 4000 + \frac{14000}{23} \approx 4609$

Answer: mode is about 4600 runs

2 Reading an ogive

7. An ogive for the monthly salaries of the workers in a factory passes through the points below. Find (a) the total number of workers (b) the median class (c) the lower-quartile class (d) the number of workers earning at least ₹6,000 but less than ₹10,000. ICSE 2025, 4 marks

Salary less than (₹)	2000	4000	6000	8000	10000	12000
Cumulative number of workers	10	20	35	55	70	75

SOLUTION

(a) The last cumulative frequency is the total. **75 workers.**

(b) $\frac{N}{2} = 37.5$. The cumulative frequency passes 37.5 between 6000 (35) and 8000 (55). **Median class: 6000–8000.**

(c) $\frac{N}{4} = 18.75$. The cumulative frequency passes 18.75 between 2000 (10) and 4000 (20). **Lower-quartile class: 2000–4000.**

(d) Workers below 10000 minus workers below 6000 $= 70 - 35$. **35 workers.**

8. The ages of 100 policy holders are given below. Draw an ogive and use it to find (a) the median age (b) the number of policy holders older than 52 years. ICSE 2024, 4 marks

Age (years)	20–25	25–30	30–35	35–40	40–45	45–50	50–55	55–60
Policy holders	2	4	12	20	28	22	8	4
Cumulative frequency	2	6	18	38	66	88	96	100

SOLUTION

Plot the points $(25, 2)$, $(30, 6)$, $(35, 18)$, $(40, 38)$, $(45, 66)$, $(50, 88)$, $(55, 96)$, $(60, 100)$, start the curve from $(20, 0)$ and join them with a smooth curve.

(a) $\frac{N}{2} = 50$. From 50 on the vertical axis go across to the curve and down to the age axis: about 42 years.

(b) From 52 on the age axis go up to the curve and across: about 91 policy holders are 52 or younger. So about $100 - 91 = 9$ are older.

Answer: (a) median ≈ 42 years (b) about 9 policy holders (readings from a graph can differ by one unit)

9. The weights of 60 students are given below. Draw an ogive and use it to find (a) the median (b) the upper quartile (c) the number of students who weigh more than 37 kg. ICSE 2023, 6 marks

Weight (kg)	28–30	30–32	32–34	34–36	36–38	38–40	40–42	42–44
Number of students	2	4	10	13	15	9	5	2

SOLUTION

Cumulative frequencies: 2, 6, 16, 29, 44, 53, 58, 60. Plot them against the upper limits 30, 32, ..., 44, start the curve at $(28, 0)$ and join the points smoothly.

(a) $\frac{N}{2} = 30$: reading across from 30 gives about 36.1 kg.

(b) $\frac{3N}{4} = 45$: reading across from 45 gives about 38.2 kg.

(c) Reading up from 37 kg gives about 37 students at or below it, so about $60 - 37 = 23$ weigh more.

Answer: (a) about 36.1 kg (b) about 38.2 kg (c) about 23 students (readings from a graph can differ slightly)

10. The distances thrown by 40 students in a shot-put competition are given below. Draw an ogive and use it to find (a) the median (b) the upper quartile (c) the number of students who threw more than 16.5 m. ICSE 2020, 6 marks

Distance (m)	12–13	13–14	14–15	15–16	16–17	17–18	18–19
Number of students	3	9	12	9	4	2	1

SOLUTION

Cumulative frequencies: 3, 12, 24, 33, 37, 39, 40. Plot them against the upper limits 13, 14, ..., 19, start the curve at $(12, 0)$ and join the points smoothly.

(a) $\frac{N}{2} = 20$: reading across from 20 gives about 14.7 m.

(b) $\frac{3N}{4} = 30$: reading across from 30 gives about 15.7 m.

(c) Reading up from 16.5 m gives about 35 students, so about $40 - 35 = 5$ threw further.

Answer: (a) about 14.7 m (b) about 15.7 m (c) about 5 students

11. The marks of 120 students in a test are given below. Draw an ogive and use it to estimate (a) the median (b) the number of students who failed if the pass mark is 50 (c) the upper quartile.

ICSE 2019, 6 marks

Marks	0–10	10–20	20–30	30–40	40–50	50–60	60–70	70–80	80–90	90–100
Students	5	9	16	22	26	18	11	6	4	3

SOLUTION

Cumulative frequencies: 5, 14, 30, 52, 78, 96, 107, 113, 117, 120. Plot them against the upper limits 10, 20, ..., 100, start the curve at $(0, 0)$ and join the points smoothly.

(a) $\frac{N}{2} = 60$: reading across from 60 gives about 43 marks.

(b) Reading up from 50 marks gives 78, so 78 students scored below 50.

(c) $\frac{3N}{4} = 90$: reading across from 90 gives about 57 marks.

Answer: (a) about 43 (b) 78 students (c) about 57

12. The heights of 60 boys are given below. Draw an ogive and use it to estimate (a) the median (b) the lower quartile (c) the number of boys taller than 158 cm. ICSE 2018, 6 marks

Height (cm)	135–140	140–145	145–150	150–155	155–160	160–165	165–170
Number of boys	4	8	20	14	7	6	1

SOLUTION

Cumulative frequencies: 4, 12, 32, 46, 53, 59, 60. Plot them against the upper limits 140, 145, ..., 170, start the curve at $(135, 0)$ and join the points smoothly.

(a) $\frac{N}{2} = 30$: reading across from 30 gives about 149.5 cm.

(b) $\frac{N}{4} = 15$: reading across from 15 gives about 146 cm.

(c) Reading up from 158 cm gives about 50 boys, so about $60 - 50 = 10$ are taller.

Answer: (a) about 149.5 cm (b) about 146 cm (c) about 10 boys

13. The daily wages of 80 workers are given below. Draw an ogive and use it to estimate (a) the median wage (b) the lower quartile (c) the number of workers who earn more than ₹625 a day.

ICSE 2017, 6 marks

Wages (₹)	400–450	450–500	500–550	550–600	600–650	650–700	700–750
Number of workers	2	6	12	18	24	13	5

SOLUTION

Cumulative frequencies: 2, 8, 20, 38, 62, 75, 80. Plot them against the upper limits 450, 500, ..., 750, start the curve at $(400, 0)$ and join the points smoothly.

(a) $\frac{N}{2} = 40$: reading across from 40 gives about ₹604.

(b) $\frac{N}{4} = 20$: reading across from 20 gives ₹550.

(c) Reading up from ₹625 gives about 50 workers, so about $80 - 50 = 30$ earn more.

Answer: (a) about ₹604 (b) ₹550 (c) about 30 workers

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